TanitAD-4B: Hierarchical Latent World Models with Typed Abstraction Interfaces for Data-Efficient Autonomous Driving

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Abstract

End-to-end driving models have won the architecture argument, yet they remain data-hungry, opaque, unaware of their own limits and expensive at inference time. We present *TanitAD-4B*, a programme that attacks all four weaknesses with one design principle: *drive at every level of abstraction, and make every level accountable*. The 4B architecture factorises driving into four brains (operative, tactical, strategic and fallback), each with its own clock, horizon, state space and predictor. The operative level owns up to 2 s, the tactical level up to 6 s and the strategic level 6 to 30 s. Every level reads the current measured ego state. The tactical level emits goals and constraints that act as the action conditioning the operative level, and the output is one trajectory or several, up to 6 s. Goals condition strictly downward and latents flow upward through stop-gradient barriers. We describe the self-supervised world-model objective behind the programme (residual multi-horizon latent prediction, provable anti-collapse via sliced Epps–Pulley regularisation, inverse dynamics and calibrated imagination of unobserved areas) and its current reference implementation, *v8F-mini*. v8F-mini makes the hierarchy *typed*: route checkpoints, a 60-dimensional tactical code of goals, manoeuvres and continuous constraints, and a trajectory that is an exact residual on a frozen learned kinematic prior. With only 7.2 M trainable parameters, trained in four hours on one 8 GB consumer GPU, it reduces 6 s displacement error against its learned kinematic floor by 0.31/0.43 m at two training seeds in open loop. A zero-training post-map makes it better than that floor on *every* row of a pre-registered metric set. Driving on its own executed motion, the adopted model lowers 6 s error by 1.26/1.19 m and improves at least one row in each of the four metric families: longitudinal, lateral, tactical and strategic (a measurement; the strategic bars are paused by decision). Split by horizon band, on-policy (DAgger) data carries the 2–6 s band and an acceleration-matching term the 0–2 s band; a per-band loss with the operative level detached from the tactical one adds a reproduced 0–2 s gain, and the tactical level's top three manoeuvres contain the logged one in 90 % of windows. Oracle reads place 87–89 % of the remaining tactical headroom in the tactical level's continuous constraints, chiefly target speed, and its sharpest cause in missed stop decisions: the operative level executes a correct stop (1.5 m lower tactical error on stop-point windows), while loss re-weighting only moves the stop decision along one ROC curve, so the next lever is stop perception. All of it rests on an evaluation doctrine built for decisions rather than leaderboards: pre-registered bars with both outcomes committed, episode-cluster bootstrap intervals, explicit seed variance and four metric families on every evaluation. We close with the remaining frontier (assertive launch from standstill), the re-rendered simulation tier built to address it, and a research roadmap toward a 4B driver within a budget of at most 800 M parameters, deployable within 100 ms on an edge computer.

1 Introduction

Driving is a hierarchy of decisions on separated time scales. A route is chosen over tens of seconds, a manoeuvre with its parameters over seconds, a control profile over tenths of a second. Monolithic end-to-end planners collapse this structure into one mapping from pixels to waypoints. That buys simplicity and gives up four things a production driver needs:

- **data efficiency**, because every situation must be learned jointly at every scale;
- **inspectability**, because no intermediate decision is exposed;
- **self-knowledge**, because nothing in the model says when it is outside its competence;
- **efficient inference**, because the full network runs at the control rate.

TanitAD is a research programme built on the opposite bet. A driver should *imagine* the consequences of its options in a learned latent world model, at the level of abstraction where each decision lives, and expose each level's decision as a typed, measurable interface. We call the architecture *4B*, for its four brains:

- an **operative** brain for continuous control and the trajectory, up to 6 s;
- a **tactical** brain that emits goals and constraints up to 6 s;
- a **strategic** brain for route checkpoints, detected context and the goals derived from them, over 6 to 30 s;
- a **fallback** brain that monitors the others and owns the minimal-risk manoeuvre.

Contributions.

(i) **The 4B formalism** (Section 3): per-level predictors in shrinking state spaces at slowing clocks, downward goal conditioning, and a gradient-isolation matrix that is *checked* rather than assumed.

(ii) **v8F-mini** (Section 4), a typed three-level instantiation over frozen foundation features:
  - a tactical code with 22 goals, 12 manoeuvre classes and 26 continuous constraints;
  - a trajectory parameterisation in which a zero residual reproduces a learned kinematic prior *exactly*, so every learned metre is measured against the floor.

(iii) **Mathematical analysis** (Section 5): a Lipschitz decomposition of plan error into an oracle term and propagated interface errors, the convex response of the operative layer to tactical accuracy, why whole-code teacher forcing leaves partial codes unsupported, and a three-variance account of evaluation intervals.

(iv) **Results under a decision-grade doctrine** (Sections 6 and 7): improvements over the kinematic floor in open loop and on the model's own executed motion, with at least one improved row in each of the four metric families.

(v) **A roadmap** (Sections 8 and 9): re-rendered simulation on neural reconstructions, perception adaptation, and scaling toward the 4B budget of at most 800 M parameters.

2 Related work

Hierarchical latent world models.
- H-JEPA [1] stacks action-conditioned joint-embedding predictive models that predict further ahead in coarser latents, with top-down sub-goal planning.
- Hierarchical planning with latent world models [2] reports large gains of two-level sub-goals over flat planners at matched capacity.
- In driving, Orbis 2 [3], Drive-HWM [4] and S2Planner [5] apply hierarchical or multi-scale world models on NAVSIM [23].

TanitAD differs in making the hierarchy a *supervised typed interface* whose usage is measured by intervention.

Self-supervised representations.
- DINOv3 [6] supplies frozen visual features.
- V-JEPA 2 [7] and LeJEPA [8] motivate latent prediction with isotropic-Gaussian regularisation.
- LoRA [9] is the adaptation route of our perception outlook.

Why modularity can help. Shalev-Shwartz and Shashua [10] show a sample-complexity separation in favour of semantic-abstraction training, under an assumption of approximately independent module failures, which our interface analysis addresses directly.

Exposure bias and on-policy data. Scheduled sampling [11], DAgger [12], DART [13], causal confusion [14] and the limits of behaviour cloning [15] frame the mismatch between a consumer trained on ground truth and one fed its predecessor's output.

End-to-end planners.
- Representative designs: conditional imitation [16], ChauffeurNet [17], TransFuser [18], VAD [19], Hydra-MDP [20], DiffusionDrive [21] and residual planners [22].
- They are evaluated on nuPlan [24] and NAVSIM [23].

3 The 4B architecture

3.1 Levels, clocks and predictors

Let $\ell \in \{O, T, S\}$ index the operative, tactical and strategic levels. Each level owns:
- a state $z_\ell \in \mathbb{R}^{d_\ell}$;
- a clock τ_ℓ and a horizon;
- an action space a_ℓ;
- a predictor P_ℓ in its own space.

The operative state is a spatial-grid readout of the encoder, $z_O = r(f_\theta(x))$ (never a global pool). Upper states are adapters over *stopped* lower latents, $z_T = \phi_T(\mathrm{sg}[z_O])$ and $z_S = \phi_S(\mathrm{sg}[z_T])$. Each level's dynamics are conditioned on that level's own action:

$$\begin{aligned} \hat{z}_S(t{+}K) &= P_S\big(z_S(t), a_S(t)\big), \\ \hat{z}_T(t{+}k) &= P_T\big(z_T(t), a_T(t)\big), \\ \hat{z}_O(t{+}j) &= P_O\big(z_O(t), (a,\kappa)_t\big). \end{aligned} \tag{1}$$

Goals do not enter the predictors. They condition the lower levels through the readers and the planner cost (Section 3.2). Two structural properties follow by construction and are then verified.

(1) Abstraction by down-sampling under a shrinking state. With $d_S < d_T < d_O$ and $\tau_S > \tau_T > \tau_O$, an upper level cannot represent lower-level detail even if its objective would reward it. At 0.2 Hz a 15 s strategic roll is three steps of a 256-dimensional model (about seven at 0.5 Hz), so only the operative path runs at the full rate.

(2) The gradient-isolation matrix. Let $G[u, v] = 1$ iff a gradient path from module u into module v is permitted. With $\mathcal{P}$ the planner and goal heads and $\mathcal{E}$ the encoder,

$$G[\mathcal{P}, \mathcal{E}] = 0, \quad G[\ell', \ell] = 0 \ \ (\ell' \text{ above } \ell), \quad G[\ell, \ell] = 1. \tag{2}$$

Every build checks it by asserting zero gradient on forbidden paths, with a deliberate-regression arm that must fail.

3.2 Time bands, the current ego state and the tactical action

Let $\Delta = 0.2$ s be the grid step and t the present. The levels own the bands

$$B_O = (0, 2]\ \text{s}, \quad B_T = (2, 6]\ \text{s}, \quad B_S = (6, 30]\ \text{s}. \tag{3}$$

The output of the stack is one trajectory or several, up to 6 s,

$$X_t = \big(x_{t+j\Delta}\big)_{j=1}^{30} \in \mathbb{R}^{30\times 2}, \tag{4}$$

so knots 1–10 lie in B_O and knots 11–30 in B_T. A horizon of 0.8 s is not a long-term abstract level: a level's reach has to match its band.

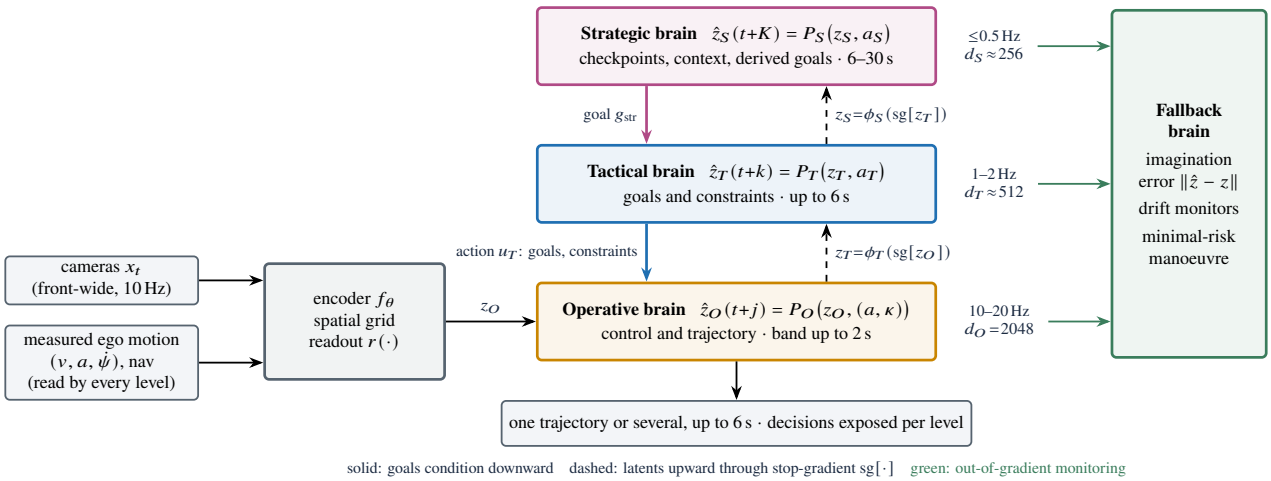


Figure 1. The 4B architecture. Four brains at separated clocks and horizons. Each predictive level owns its own predictor in its own latent space, and the state shrinks while the clock slows going up ($d_S < d_T < d_O$, $\tau_S > \tau_T > \tau_O$), so upper levels are forced to abstract. Goals condition strictly downward, through the readers and the planner cost; the predictors themselves are action-conditioned and goal-free. Latents pass upward only through stop-gradient adapters, and the gradient-isolation matrix of Eq. (2) is tested on every build. The fallback brain consumes every level's imagination error and owns the minimal-risk manoeuvre.

The cascade. Let e_t be the current measured ego state, including yaw rate, and V_t the vision tokens of the current and past frames, with the navigation command and speed limit. Every level reads e_t, and no level reads a state planned by another level:

$$\begin{aligned} s_t &= f_S(e_t, V_t), \\ u_t &= f_T(e_t, V_t, \hat{s}), \\ X_t &= f_O(e_t, V_t, \hat{s}, \hat{u}), \end{aligned} \tag{5}$$

where a hat marks a held output re-expressed in the current ego frame by measured odometry. The tactical output is a *tactical action*,

$$u_t = (g_t, \lambda_t^{\mathrm{lat}}, \lambda_t^{\mathrm{lon}}, \gamma_t), \tag{6}$$

the goals, the lateral and longitudinal classes and the constraints, over a horizon of up to 6 s. T has no trajectory head. u_t conditions O as its action, so the plan is a function of the context *given* u_t. A single decoder produces all 30 knots, which makes the plan continuous across 2 s by construction, and the two bands differ only in the loss:

$$\mathcal{L}_O = w_{\mathrm{op}}\, \ell\big(X^{1:10}, X^{*\,1:10}\big) + w_{\mathrm{tac}}\, \ell\big(X^{11:30}, X^{*\,11:30}\big). \tag{7}$$

For M hypotheses, $X_t^{(m)} = f_O(\cdot \mid \hat{u}^{(m)})$ with $\hat{u}^{(m)}$ the M best tactical actions; $M = 1$ is the arm of record.

The strategic level. Its output is $\sigma_t = (C_t, p_t, \gamma_t^S)$: checkpoints C_t at $\tau \in \{6, 9, \ldots, 30\}$ s; the posteriors p_t of the detected context (country, period of day, illumination, weather, visibility and road surface), read from the current and past images only; and the goals γ_t^S derived from them. σ_t enters f_T as a detached input, so strategic goals reach O through u_t. No output of a situation classifier enters any goal path.

World-model planning. The levels $l = 1, 2, 3$ step at $\tau_l = 0.2, 0.6, 3.0$ s and are trained to reaches of 2, 6 and 18 s. Following H-JEPA [1], with g a goal latent, $\tilde{z}_i^{l+1}$ the rollout of level l lifted to the level above, and $l = 2, 1$,

$$\begin{aligned} a^{3,*} &= \arg\min_{a^3} \big\| \hat{z}_{T_3}^{3} - g \big\|^2, \\ a^{l,*} &= \arg\min_{a^l} \big\| \tilde{z}_K^{l+1} - \hat{z}_K^{l+1,*} \big\|^2 \\ &\quad + \lambda \sum_{i<K} \big\| \tilde{z}_i^{l+1} - \hat{z}_i^{l+1,*} \big\|^2. \end{aligned} \tag{8}$$

Each level plans from its own encoded current state, and the only downward signal is the predicted state of the level above. This planner is the baseline against which other approaches are compared.

Status. Equation (5) is trained on v8F-mini as described in Section 4, with equal knot weights and tactical labels over 2–8 s. The band losses, the 2–6 s labels, the strategic extension and the world-model planner are built and tested, and the world models for these bands are in training. A first planner read on the earlier models, with a true-future goal (T0), did not beat holding the action. No driving result exists yet for this design.

3.3 Self-supervised world-model objective

The programme's world model is trained without perception labels, without HD maps and without reward. Its objective is

$$\begin{aligned} \mathcal{L} = \mathcal{L}_{\mathrm{pred}} + \lambda_{\mathrm{tac}}\mathcal{L}_{\mathrm{tac}} + \lambda_{\mathrm{sig}}\big(\mathcal{S}(Z) + \mathcal{S}(\hat{Z})\big) \\ + \lambda_{\mathrm{inv}}\mathcal{L}_{\mathrm{inv}} + \lambda_{\mathrm{im}}\mathcal{L}_{\mathrm{imag}}. \end{aligned} \tag{9}$$

Residual multi-horizon prediction. $\hat{z}_{t+k} = z_t + \Delta_k(z_{t-W+1:t}, a_{t-W+1:t})$, scored by a *change-weighted* error with per-dimension weights $w \propto |z_{t+k} - z_{t+k-1}|$ normalised to unit mean. On driving data only 5–11 % of a frame changes per step, so the weighting puts the gradient where the consequences are.

Provable anti-collapse. $\mathcal{S}(\cdot)$ is the sliced Epps–Pulley statistic of LeJEPA [8]. Embeddings are projected onto M

fresh unit directions, and each projection Y (a sample of size n) is tested against $\mathcal{N}(0,1)$:

$$T_{n,\beta} = \frac{1}{n}\sum_{j,k} e^{-\frac{\beta^2(Y_j-Y_k)^2}{2}} - \frac{2}{\sqrt{1+\beta^2}}\sum_{j} e^{-\frac{\beta^2 Y_j^2}{2(1+\beta^2)}} + \frac{n}{\sqrt{1+2\beta^2}}. \tag{10}$$

By the Cramér–Wold theorem, standard normal one-dimensional projections in all directions imply a joint standard Gaussian, the embedding law LeJEPA shows to be optimal; M random directions test this only approximately. The gradients stay uniformly bounded [8].

Inverse dynamics. $\mathcal{L}_{\text{inv}} = \|g(z_t, z_{t+1}) - a_t\|^2$ uses proprioception as free supervision and forces controllable state into the latent.

Imagination of unobserved areas. Masked sectors evolve by semi-Lagrangian advection under a learned flow, $\hat{z}_{t+1}(u) = z_t(u - v_\theta(u)\Delta t)$, refined by attention, with a per-cell log-variance $s(u)$:

$$\mathcal{L}_{\text{imag}} = \sum_u w(u)\left[e^{-s(u)}\frac{\|\hat{z}(u)-z(u)\|^2}{2} + \frac{s(u)}{2}\right], \tag{11}$$

with $w = 1$ on hidden cells and 0.1 on visible ones. The model must know *where it cannot know*. The variance field is the world-model-native trigger for sensor steering and for the fallback brain.

3.4 Why a hierarchy is data-efficient

Two arguments motivate the factorisation [2, 10].

(i) **Horizon.** Planning regret in learned models grows with the horizon, so a short per-level horizon has a smaller bound at that level. Summed over the levels, equal per-step errors give the undivided bound back: the gain requires the coarser levels to carry a smaller per-step error, which is an empirical question.

(ii) **Recombination.** If a driving situation is a combination of a place $p \in \mathcal{P}$, a manoeuvre $m \in \mathcal{M}$ and a control profile, a flat learner must cover $|\mathcal{P}| \cdot |\mathcal{M}|$ combinations. A hierarchy that learns each level separately covers roughly $|\mathcal{P}| + |\mathcal{M}|$, provided places and manoeuvres combine independently.

3.5 Fallback and self-knowledge

The relative imagination error $\|\hat{z}_t - z_t\|$ is a free, always-on familiarity signal. Layered monitors arm a deterministic minimal-risk manoeuvre:

- imagination error at 10 Hz;
- routing statistics at 1 Hz;
- Mahalanobis drift at 0.1 Hz.

Their logs are the substrate for the ISMR/DSSAD incident-reporting requirements: self-monitoring *is* the evidence trail.

The programme's 263 M-parameter 4B flagship was its first arm to beat the kinematic floor in open loop: ADE@2 s 0.452 m against 0.500 m for the best of three kinematic models and 0.523 m for CTRV. On windows where kinematics diverge from the realised future, it beat CTRV by 0.80 m, and the whole of that advantage came from vision.

4 v8F-mini: a typed reference implementation

4.1 Inputs and notation

Time runs on a 0.2 s grid. For a window ending at t, the model reads:

- **scene tokens** $z_{t'} \in \mathbb{R}^{40\times1024}$, frozen DINOv3-L/16 features of the cylindrical front-wide frame, pooled to a 4×10 grid, four frames per layer;
- **ego features** $f_t = (v, a, \kappa, v^2, va, v\kappa, v^2\kappa, a\kappa)$, with curvature κ from the *measured yaw rate* (never a steering angle);
- **a navigation command** n_t;
- **a speed bucket** ℓ;
- **measured odometry** between frames.

The output is a 6 s plan $\tau \in \mathbb{R}^{60\times2}$ in the ego frame, scored on its 30 knots at 0.2 s. Several plans are possible, one per top tactical hypothesis; the arm reported here emits one.

4.2 The S/T/O decomposition

S ticks at about 1 Hz, T at about 2 Hz and O at 10 Hz. On the cache grid the model uses sample-and-hold with quantised ages:

- T last ticked $a_T \in \{0, 1, 2\}$ steps ago;
- S ticked $u \le 4$ steps before T, so $a_S = a_T + u$.

With $t_T = t - a_T$ and $t_S = t - a_S$,

$$\begin{aligned}\hat{k} &= f_S\big(z_{t_S}, f_{t_S}, n_{t_S}, \ell\big),\\ \hat{c} &= f_T\big(z_{t_T}, f_{t_T}, n_{t_T}, \ell, \Pi_{t_S\to t_T}\hat{k}\big),\\ \hat{\tau} &= f_O\big(z_t, f_t, n_t, \ell, \Pi_{t_S\to t}\hat{k}, \hat{c}, (a_T, a_S)\big),\end{aligned} \tag{12}$$

where $\Pi_{a\to b}$ re-expresses held checkpoints from frame a to frame b by odometry. The conditioning sets are strictly nested, $\text{in}(S) \subset \text{in}(T) \subset \text{in}(O)$. Information flows between levels only through *held outputs*, never through shared hidden state. Each level reads the measured ego state of its own tick, and no level reads a state planned by another level. Each f_ℓ is a *Reader*: four learned queries cross-attend to the conditioning and scene tokens through a pre-norm transformer decoder, followed by an MLP head.

4.3 The tactical code as a typed interface

$$\begin{aligned}c = \big[\ &\underbrace{\sigma(g)}_{22\text{ goals}} \ \big|\ \underbrace{\text{softmax}(\lambda_{\text{lat}})}_{5} \ \big|\\ &\underbrace{\text{softmax}(\lambda_{\text{lon}})}_{7} \ \big|\ \underbrace{(\gamma-\mu)/\sigma_{\text{cons}}}_{26\text{ constraints}}\ \big] \in \mathbb{R}^{60}.\end{aligned} \tag{13}$$

The 26 constraint cells cover turn timing and geometry, longitudinal target speeds and reach times, the stop point, the lead gap and time gap, and the speed band. The code is the tactical action of Section 3.2: T emits goals and constraints, has no trajectory head, and $\hat{c}$ conditions O. The route checkpoints of this arm sit at 30, 50 and 80 m, with a median arrival of 3, 5 and 8 s, so they do not yet cover the strategic band in seconds; Section 3.2 gives the extension.

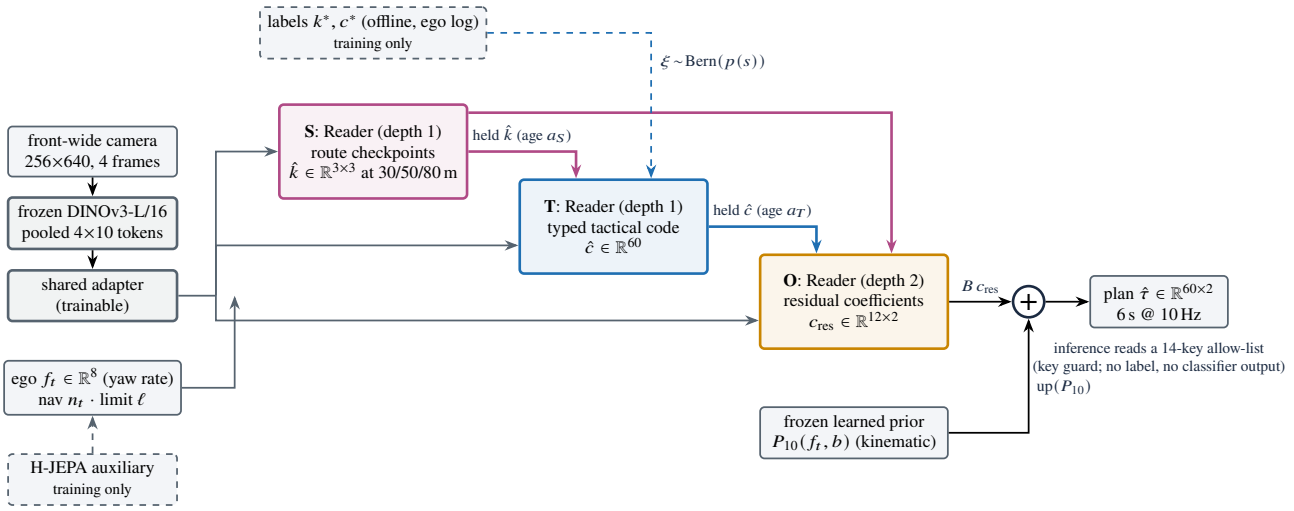


Figure 2. v8F-mini. Three learned readers over frozen DINOv3 features and a shared trainable adapter. S emits route checkpoints, T a typed 60-dimensional tactical code, and O the coefficients of a ramped cosine basis added to a *frozen* learned kinematic prior. Upper outputs are *held* with quantised ages and re-expressed in the current frame by odometry. Labels enter only through scheduled teacher forcing in training. Inference reads a fixed allow-list of inputs, enforced by a key guard.

An information-theoretic reading. Ground-truth codes $C^* = g(Y)$ are deterministic functions of the human future Y, so the data-processing inequality gives $I(C;Y) \geq I(C;C^*)$. Supervised agreement with C^* is therefore a lower-bound proxy for informativeness about the future. O also reads the scene, so the code is a *typed side channel*, not a strict bottleneck, and its usage is measurable by intervention (Section 7).

4.4 Trajectory as an exact residual on a kinematic prior

Let $P = P_{10}(f_t, b) \in \mathbb{R}^{30\times2}$ be a *frozen* learned prior: a damped base path re-timed by constant acceleration, plus an eight-feature MLP fitted once. Let up interpolate it to 0.1 s. The plan is

$$\tau = \mathrm{up}\big(P_{10}(f_t, b)\big) + B\,c_{\mathrm{res}}, \qquad B_{ik} = \tfrac{i+1}{N}\cos\Big(\tfrac{k\pi(i+\frac{1}{2})}{N}\Big), \tag{14}$$

with $N = 60$, $k = 0, \ldots, 11$, $c_{\mathrm{res}} = s \cdot \mathrm{MLP}(h_O)$, $s = 2\,\mathrm{m}$, and the last layer zero-initialised.

Proposition 1 (Floor identity, ramp and band limit)**.**

(a) ***Floor identity.*** *$c_{\mathrm{res}} = 0$ implies $\tau = \mathrm{up}(P_{10})$ exactly, so the untrained model* is *the prior.*

(b) ***Ramp.*** *$|B_{ik}| \leq (i+1)/N$, so the residual vanishes linearly as $t \to 0$ and the plan starts at the prior's origin.*

(c) ***Band limit.*** *With $k \leq 11$ the cosine factor has at most eleven half-cycles over 6 s, below 1 Hz, and the linear ramp modulates it smoothly: a smooth adjustment, while the prior carries the kinematics.*

Every learned metre is thus measured against the floor it must beat. Each property is pinned by a unit test with a mutation check.

4.5 Losses and scheduled teacher forcing

The training loss is

$$\mathcal{L} = w_{\mathrm{ck}}\mathcal{L}_{\mathrm{ck}} + w_{\mathrm{tac}}\big(\mathcal{L}_{\mathrm{goal}} + \tfrac{1}{2}(\mathcal{L}_{\mathrm{lat}} + \mathcal{L}_{\mathrm{lon}}) + w_{\mathrm{cons}}\mathcal{L}_{\mathrm{cons}}\big) + w_{\mathrm{traj}}\big(\tfrac{1}{3}\mathcal{L}_{\mathrm{ADE}} + \tfrac{1}{12}\mathcal{L}_{\mathrm{FDE}}\big) + w_{\mathrm{pen}}\textstyle\sum_r \rho_r + w_{\mathrm{wm}}\mathcal{L}_{\mathrm{wm}}. \tag{15}$$

Its terms:

- **partial-label cross-entropies** $-\log \sum_{c\in\mathcal{A}} p_c$ handle the sets of allowed manoeuvres;
- **constraint cells** use a masked Huber loss;
- **rule penalties** ρ_r (speed, stop, turn, lead) are differentiable hinges that are positive *iff* a frozen rule reports a violation;
- **the world-model auxiliary** $\mathcal{L}_{\mathrm{wm}}$ is an H-JEPA objective on the shared adapter [1].

During training, O and T read a mixture of predicted and ground-truth upper outputs,

$$\tilde{c} = \xi_c\, c^* + (1 - \xi_c)\,\hat{c}, \qquad \xi_c \sim \mathrm{Bernoulli}\big(p(s)\big), \qquad p(s) = 1 - 0.8\min(s/S, 1), \tag{16}$$

which anneals the share of ground truth from 1.0 to 0.2 [11]. At inference $\xi \equiv 0$, and labels are never read.

4.6 Admissibility by construction

- Inference reads a fixed allow-list of 14 keys. A key guard records every access during the forward pass, and a test proves that a deliberately peeking model is caught.
- The goal path may not contain the output of any situation classifier.
- Labels may use the ego future, but inference may not.

5 Mathematical background

5.1 Plug-in composition and propagated interface error

By the chain rule, $p(y \mid x) = \sum_{k,c} p(k \mid x)\, p(c \mid x, k)\, p(y \mid x, k, c)$. Factorisation alone is an identity; its value lies in *supervision structure*. Each factor is fitted against its own offline target, and inference is a plug-in composition $\hat{y} = \mu_O(x, \hat{k}, \hat{c})$.

Proposition 2 (Error decomposition)**.** *If μ_O is L_k- and L_c-Lipschitz in its code arguments, then*

$$\|\hat{y} - y\| \le \underbrace{\|\mu_O(x, k^*, c^*) - y\|}_{\text{oracle term}} + L_c\|\hat{c} - c^*\| + L_k\|\hat{k} - k^*\|. \tag{17}$$

The oracle term is measurable by substituting ground-truth codes. The gap between the model's own code and the oracle is the propagated interface error, and it is the headroom the hierarchy offers (Section 7).

5.2 Convex response to tactical accuracy

Define the partial substitution $\hat{c}_\alpha = \hat{c} + \alpha(c^* - \hat{c})$ on the 26 constraint cells, and the recovered fraction

$$r(\alpha) = \frac{\mathrm{ADE}(\hat{c}) - \mathrm{ADE}(\hat{c}_\alpha)}{\mathrm{ADE}(\hat{c}) - \mathrm{ADE}(\hat{c}_1)}. \tag{18}$$

Linear response gives $r(\alpha) = \alpha$. We measure $r(0.25) = 0.08$–0.16, $r(0.5) = 0.20$–0.36 and $r(0.75) = 0.48$–0.73. The response is *convex*: O is flat near its own code and steep near the truth. The value of tactical accuracy therefore compounds, and the last metres of tactical precision buy the most driving.

5.3 Interface support and exposure bias

Whole-code teacher forcing trains O on the two-point mixture $p\,\delta_{c^*(x)} + (1-p)\,\delta_{\hat{c}(x)}$. A partial code $c_u = (c^*_u, \hat{c}_{\bar{u}})$ lies off that support, so its response is unconstrained by the training loss. The remedy is to widen the support: draw an independent ξ_u per unit of the code at the same marginal $p(s)$,

$$\tilde{c}_u = \xi_u c^*_u + (1 - \xi_u)\hat{c}_u, \qquad \xi_u \overset{\text{iid}}{\sim} \mathrm{Bern}\big(p(s)\big). \tag{19}$$

This keeps each cell's marginal unchanged while covering 2^7 corners per window instead of 2.

The same logic applies along time. A policy trained only on expert states incurs compounding error of order εT^2, while on-policy aggregation (DAgger [12]) reduces it to order εT under that reduction's no-regret assumptions. v8F-mini realises this by replacing 10 % of the training samples with the model's *own rolled-out states*, labelled from the human future.

5.4 Estimator theory: three variances, one interval

Windows are grouped into $M = 141$ episodes. For arms a, b the point estimate is the full-set mean difference $\hat{\Delta} = \frac{1}{N}\sum_w [d_a(w) - d_b(w)]$. The paired episode-cluster bootstrap resamples episodes with replacement, using the same draw for both arms, and reports percentile intervals. By the law of total variance, with training randomness θ, inference randomness ω and the episode draw E,

$$\mathrm{Var}(\hat{\Delta}) = \underbrace{\mathrm{Var}_\theta\, \mathbb{E}[\hat{\Delta} \mid \theta]}_{\text{training seeds}} + \underbrace{\mathbb{E}_\theta \mathrm{Var}_\omega\, \mathbb{E}_E[\hat{\Delta} \mid \theta, \omega]}_{\text{inference}} + \underbrace{\mathbb{E}_{\theta,\omega}\mathrm{Var}_E[\hat{\Delta} \mid \theta, \omega]}_{\text{episodes}}. \tag{20}$$

The bootstrap estimates the episode term only. TanitAD therefore reports the seed term separately, through two training seeds and a pre-registered floor $F = |\Delta_{s_0} - \Delta_{s_1}|$ with bar $2F$. For a deterministic forward pass the inference term is exactly zero, and the programme verifies that on re-evaluation (maximum per-window deviation 0.0 m).

6 Evaluation doctrine

Tiers. Every number carries its tier.

- T0: teacher-forced or oracle-conditioned. A diagnostic, never driving performance.
- Open loop: the model's own held outputs, with logged ego data.
- T1-kin: the model's *own executed motion* rebuilds the ego state at every 2 s replan (three replans of 10 executed knots per window) while the cameras stay logged.
- T2: cameras re-rendered at the car's own pose and time, with replayed traffic, Section 8.

Four metric families, never one score. Every evaluation reports:

- *longitudinal*: speed, along-track, acceleration and distance keeping;
- *lateral*: heading, yaw rate, curvature and cross-track;
- *tactical*: manoeuvre-decision agreement;
- *strategic*: route-checkpoint error.

The families are reported per family and never pooled. The strategic bars are paused by decision until the tactical and operative layers are proven, so strategic rows here are measurements and not a pass.

Pre-registration. Each arm has a card committed before its data. The card fixes the primary bar, the mechanism read, the refusals and *both* outcomes with their next steps. Every training run is bound by a launch gate to its commit, argv and data hashes, and every check carries a deliberate-regression arm that must fail.

Data. Training uses 4,512 PhysicalAI-AV clips (306,566 windows). Evaluation uses 141 held-out episodes (2,399 windows), disjoint by clip and by recording session.

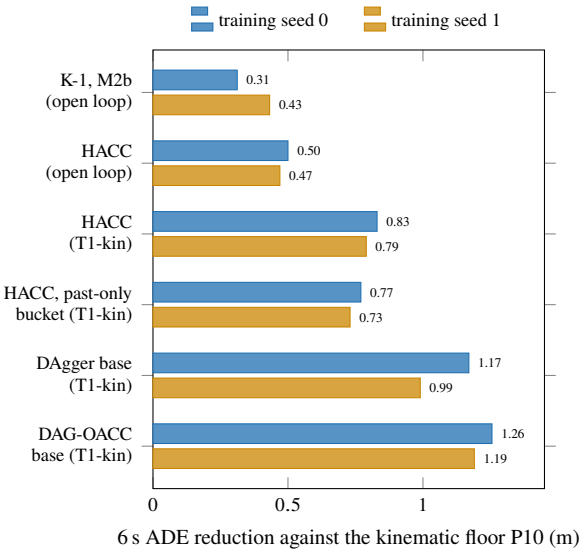


Figure 3. Six-second error reduction against the learned kinematic floor, at two training seeds. Every bar is a paired episode-cluster bootstrap contrast on 141 held-out episodes, and every one is separated from zero. The tier is given per bar.

7 Results

The hierarchy beats its floor. Against the learned kinematic floor P10, which is exactly the model's own output at a zero residual by Eq. (14), v8F-mini (M2b) lowers 6 s error by 0.312/0.432 m at two seeds, beyond a pre-registered bar of twice the seed floor (Fig. 3, Table 1). A zero-training post-map (HACC) blends O's plan with its own prior through per-step weights fitted on training windows with an acceleration term. With it, the model is separated-better than P10 on every row of a pre-registered set, including 2 s error (−0.011/−0.008 m), 6 s error (−0.50/−0.47 m) and acceleration (−0.022/−0.040 m/s^2), and separated-worse on none.

The margin survives the model's own motion. On T1-kin, where the ego state is rebuilt from the model's own executed motion at each replan, the HACC model beats P10 by 0.83/0.79 m. Two stress versions keep most of the margin:

- a speed bucket rebuilt from the last 4 s of measured speed alone keeps 0.77/0.73 m;
- freezing the camera at the first frame for 6 s as well keeps 0.73/0.64 m.

The margin therefore rests on the model's reading of the scene and of its own state, not on future information.

On-policy data makes the tactical brain robust. Replacing 10 % of training samples by the model's own rolled-out states (DAgger, $p = 0.1$) raises the tactical brain's lateral macro-F1 by 0.108/0.118 and its longitudinal macro-F1 by 0.055/0.081 on the model's own motion. It also lowers executed 6 s error by 0.54/0.40 m. It is the first training lever to clear a T1 bar on both heads. The adopted base built on it (DAG-OACC) lowers 6 s error against P10 by 1.26/1.19 m on its own motion. At least one row improves in each of the four families (longitudinal, lateral, tactical and strategic).

Every level contributes. The strategic brain's route checkpoints beat a kinematic arc at every horizon, by up to 8.8 m at 80 m. The tactical interface is used, as substituting the ground-truth code cuts 6 s error from about 2.1 to about 1.0 m. Proposition 2 reads this as roughly 1.1 m of headroom available through better tactical predictions. Because the response is convex, that headroom concentrates in the final gains in tactical precision.

Table 1. Headline results (MEASURED; paired episode-cluster bootstrap, $B = 2000$, 141 episodes; values at seed 0 / seed 1).

Result	Tier
6 s ADE −0.312 / −0.432 m vs P10, beyond the pre-registered $2F = 0.2415$ m (12.5–17.4 % lower error)	open loop
HACC post-map: better than P10 on *every* row of the pre-registered set (ADE2, ADE6, accel., curvature, speed, heading, yaw rate, decisions, consistency)	open loop
6 s ADE −0.83 / −0.79 m vs P10 on own executed motion; −0.77 / −0.73 m with a past-only speed bucket	T1-kin
Route checkpoints beat a kinematic arc by 1.2/3.3/7.8 m and 1.3/3.5/8.8 m at 30/50/80 m	T1-kin
DAgger: tactical macro-F1 +0.108/+0.118 (lateral), +0.055/+0.081 (longitudinal); 6 s ADE −0.54/−0.40 m vs its control	T1-kin
Adopted base: 6 s ADE −1.26 / −1.19 m vs P10; every family has a separated-better row	T1-kin
Per band: DAgger rows lower 2–6 s ADE by 0.82 / 0.62 m; the acceleration term lowers 0–2 s ADE by 0.062 / 0.079 m and 0–2 s acceleration error by 0.97 / 1.14 m/s^2	T1-kin
Band loss, operative level detached: 0–2 s ADE −0.017 / −0.015 m beyond a same-recipe replicate; no 2–6 s row moves	T1-kin
Top-3 tactical manoeuvres contain the logged one in 90 % of windows (top-1: 57–59 %); best of three plans −0.35 to −0.39 m 2–6 s ADE (oracle)	T1-kin
Tactical headroom: 87 / 89 % in the continuous constraints; a forced correct stop lowers stop-point tactical ADE by 1.54 / 1.48 m; stop re-weighting moves recall only along one ROC curve (oracle)	T1-kin
Ground-truth tactical code: 6 s ADE from ≈2.1 to ≈1.0 m (interface headroom)	T0

Which lever carries which band. Under the band definition of Section 3.2, we retrained the operative and tactical levels with a per-band loss and with the operative level's loss detached from the tactical level, at two seeds and with a same-recipe replicate (T1-KIN throughout). The 0–2 s plan improves at both seeds beyond the replicate: ADE by 0.017/0.015 m, about 6 %, and FDE at 2 s by 0.039/0.027 m. No row of the 2–6 s band moves, and the replicate says why: two runs of the same recipe differ there by 0.29 m. A ladder of banked fine-tunes splits the adopted base into its three changes. The DAgger rows carry the tactical band (2–6 s ADE −0.82/−0.62 m, heading −4.6/−2.6°). The acceleration-matching term carries the operative band and acceleration (0–2 s ADE −0.062/−0.079 m; acceleration error −0.97/−1.14 m/s^2 at 0–2 s and −2.0/−2.6 m/s^2 at 2–6 s). Decoding the operative level once per top-three tactical manoeuvre, the logged manoeuvre is among the three in 90 % of windows against 57–59 % for the first choice, and the best

of the three plans would lower 2–6 s ADE by 0.35–0.39 m. That oracle gap is the measured headroom of a trained selection over the tactical codes, and the next lever.

Where the tactical headroom sits. We measured that lever before building it, with zero-training oracle reads on the adopted recipe (T1-KIN, first tick, both seeds, every control at its known value; the strategic family paused). A perfect selector over the top three manoeuvres captures only 24/23 % of the best-of-three gain, so the choice among manoeuvres is not where the headroom lies. Writing the ground-truth tactical code in place of the model's own lowers tactical ADE by 1.63/1.77 m. Leaving out its continuous constraints forfeits 87/89 % of that gain and leaving out the discrete decision 11/9 %; target speed is the costliest single constraint. The speed error spreads over every stratum, but stop-point windows hold 20 % of its squared mass on 5 % of the windows. There the decision misses: the goal head says stop in 10/13 % of the 117 stop-point windows, although its stop score separates them from the rest with AUROC 0.84/0.86. Forcing the correct stop into the tactical code lowers their tactical ADE by 1.54/1.48 m (30–31 %), so the operative level executes a stop once it is decided.

Three pre-registered levers then acted on the decision alone. Hardening the model's own decision at inference recovers 3–6 % of that gain at a cost on the other windows, and a threshold chosen on training clips does not help. Reweighting the stop classes at two doses raises stop recall from 0.06–0.15 to 0.45–0.62 and the false-stop rate from 0.5–1.3 % to 12–14.5 %, with AUROC unchanged within its replicate floor: the weights slide the operating point along one ROC curve and add no discrimination, and stop-point ADE does not improve beyond its 0.70 m floor. The missing piece is discrimination, which is a perception question. Its ceiling was fixed before any training: a correct stop on the 82 of 117 stop-point windows that the model already ranks highest lowers tactical ADE there by 0.96/0.86 m, beyond the floor at both seeds, with a separated lateral cost (yaw-rate error +2.2°/s) that comes with the forced stop. Linear and small MLP probes on the frozen features rank stops no better than the model (AUROC 0.84–0.87 for every input set, a 4× finer token grid included), so the first perception arm trains instead of probing: a vision-only stop-location auxiliary loss, queued to train at two seeds against same-seed controls and two replicates. Two recipe levers closed on the way: on the DAgger base, detaching the operative level costs nothing, and the band loss's operative gain shrinks to 1–2 mm and does not settle over four seeds.

8 The frontier: assertive launch, and re-rendered simulation

With the feasibility post-step, the adopted composite meets every row of the 19-row pre-registered set but one: the *launch-crawl* share. These are situations where the human pulls away from near-standstill above 5.2 m/s and the model starts more cautiously. Every zero-training lever available on CPU has been measured against this row:

- decode levers;
- a level-from-image head;
- four doses of an external progress floor.

Even with the exact human speed supplied as a ceiling diagnostic, a bounded floor closes about half of the gap.

The analysis points to a property of the tier rather than of the planner. Logged cameras cannot show a lagging car its own consequences: the lead vehicle never pulls away on screen, and the scene never moves at the car's own pace.

T2 re-perception. The re-rendered evaluation tier runs on neural reconstructions of real drives:

- 62 scenes (42 launch, 20 guard), selected by a rule frozen *before* any log was read, and staged (35.7 GB);
- a harness that reproduces the T1-kin plans bit for bit from cached features, with an in-loop encoder that reproduces the cached tokens to 2.6 % relative L_2 (cosine 0.99965).

The pre-registered design runs two conditions with the same renderer: cameras rendered at the car's *own* pose and time, and at the *logged* pose re-indexed by the car's progress. Their contrast measures the camera-clock effect directly, free of the render domain gap, and both outcomes and their next levers are committed in advance. First reads now exist on 48 scenes with replayed traffic. An encoder adapted to the render gap improves speed error at both seeds, does not separate 6 s error from the unadapted model at either seed, and is not better than the floor on every row, so the launch row stays open.

9 Outlook and research roadmap

The roadmap (Table 2) follows from the measurements. The architecture is validated at small scale on a consumer GPU, every level carries measurable value, and the remaining behaviour sits where logged-camera evaluation cannot see. Three investments turn the validated design into a driver:

- a **re-rendered simulation tier**, with first reads in hand;
- **perception adaptation** for motion and stop cues;
- **scale in data and model size** toward the 4B budget.

Because the floor identity holds at every scale, and every lever is pre-registered with deliberate-regression controls, each investment yields a measured answer rather than an anecdote.

10 Conclusion

TanitAD-4B drives at every level of abstraction and holds every level accountable. Its reference implementation shows that a typed hierarchy over frozen foundation features clears a learned kinematic floor:

- in open loop, by 0.31–0.43 m of 6 s error;
- on every row of a pre-registered set, with a zero-training post-map;
- on its own executed motion, by more than a metre, with at least one improved row in each of the four metric families.

It does so with 7.2 M trainable parameters and four GPU-hours. The mathematics explains where the remaining headroom sits: in tactical precision, with convex returns, and in the evaluation tier. The roadmap names the investments that

will convert the architecture into a deployable, self-aware driver.

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Table 2. Roadmap (*outlook*: not yet implemented unless marked).

Item	What it delivers
T2 re-perception *(first reads, 48 scenes)*	a decision-grade read of the launch row on re-rendered scenes; the longitudinal tier for every later arm
World-model planner *(built)*	top-down planning over three latent levels, the baseline for comparing other approaches; its first gates on the M2b latents do not yet beat holding the logged action (T0); giving its predictor the measured speed does not raise its skill beyond the seed floor (T0); a slow-feature basis is being read
Band retraining *(read, four seeds)*	on the DAgger base the detach costs nothing; the band loss's 1–2 mm operative gain does not settle, so it is closed as a recipe lever
Tactical selection *(oracle read)*	a perfect top-3 selector captures 24/23 % of the best-of-three headroom; the headroom sits in the speed constraint, so no selector is built
Stop perception *(queued, two seeds)*	a vision-only stop-location auxiliary on the tactical level; measured ceiling 0.86–0.96 m tactical ADE on stop-point windows (oracle)
Strategic context labels *(in production)*	per-frame country, period of day, illumination, weather, visibility, road surface
Perception adaptation *(in training, seed 0)*	LoRA on the visual encoder for motion cues; about 184 h per seed on the 20 GB pod GPU at the measured 34.5 s per step
Scale data	from 4,512 clips toward the full corpus; on-policy DAgger data from T2 rollouts
Scale size	from 7 M toward the 4B budget of at most 800 M, with each level's predictor in its own latent
Strategic topology	routing over a latent transition graph, with map structure from simulation or an external corpus
Fallback brain	imagination-error monitoring, the minimal-risk manoeuvre and ISMR/DSSAD logging
Occlusion imagination	multi-step, calibrated belief roll-out of hidden sectors
Edge deployment	Jetson Thor, at most 100 ms end to end on 4 cameras
Benchmarks	NAVSIM navtest and pre-registered T2 reads